

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

Relevant Coursework: Stochastic Processes, Probability and Statistics, Real Analysis, Data Analytics, Database Systems, Artificial Intelligence

TECHNICAL SKILLS

Analytics/Modeling: Credit risk, predictive modeling, statistical analysis, feature engineering, model validation, backtesting, ROC/AUC

Languages/Tools: Python, SQL, R, MATLAB, SAS, Power BI, Git, Excel (reconciliation, validation, reporting)

Data Platforms: SQL Server, Oracle, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker

Financial Applications: Credit risk modeling, fraud analytics, transaction monitoring, SEC filing analysis, customer risk scoring

RISK AND FINANCIAL ANALYTICS EXPERIENCE

Enterprise Risk Analyst Intern

Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI

May 2025 – Dec. 2025

Dallas, TX

- Developed an end-to-end credit risk analytics pipeline integrating PayNet inquiries and internal loan servicing data through SQL Server ETL, credit-feature engineering, Python modeling, and validation.
- Engineered borrower, contract, and asset-level features including loan-to-value ratio, account tenure, inquiry patterns, asset segments, and loan servicing attributes.
- Validated model performance with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk case identification by approximately **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing credit scorecard segmentation updates.
- Automated monthly model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by 10% and supporting recurring portfolio risk monitoring.

Financial Data Development Engineer – Fraud Detection

Caitong Securities Ltd | SQL, Oracle, Stored Procedures

Jul. 2024 – Aug. 2024

Hangzhou, China

- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% in transaction-monitoring workflows.
- Developed a customer risk scoring engine by integrating transactional and behavioral data streams for fraud analytics, reviewer triage, and AML-related review workflows.

FinChat – AI-Powered Financial Analysis System

Southern Methodist University | Python, FinGPT, LangChain, CrewAI

Jan. 2023 – May 2024

Dallas, TX (Remote Hybrid)

- Built ETL pipelines to extract and process 100+ SEC filings from EDGAR, structuring unstructured financial data into a scalable data lake for metric extraction and trend detection.
- Implemented LangChain and CrewAI workflows for contextual financial interpretation, improving filing-analysis accuracy by 3% and reducing manual review time for filing analysis.

ADDITIONAL EXPERIENCE

Software Engineering Intern – Customer Analytics

Joblogic-X Corporation | Python, APIs, Git

Nov. 2023 – Jun. 2024

Dallas, TX

- Integrated social-media and transactional order data into a customer analytics pipeline for ranking customers, segmenting engagement patterns, and supporting marketing strategy.

Researcher & TA – Biological Data Analytics

University of Texas at Austin | R, Linux, AWS, Docker

Jan. 2024 – Dec. 2024

Austin, TX

- Applied statistical modeling (differential-expression analysis, PCA, permutation-based FDR) to 250GB of RNA-seq data as a research TA, identifying 48 key genes across reproducible R/Linux pipelines.

SELECTED RESEARCH AND PUBLICATIONS

- Coauthored **5 peer-reviewed publications** applying machine learning and statistical analysis to computer vision, neuroengineering, and social-media data.
- Selected venues include *Expert Systems with Applications*, *Frontiers in Neurology*, *Frontiers in Neuroscience*, and IEEE ICIPCA.